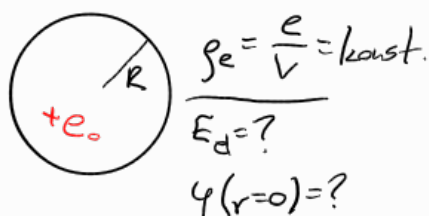


Iz meritev: prvi približek: jedro = kroglica iz nukleonov, s pribl. konst. gostoto
 • vezavna energija $\propto A$ (približno)

$\Rightarrow$ trdna središča, interakcija je kratkega doseg, $r_{\text{jedra}} \approx r_1 A^{1/3}$
 $\downarrow$
 1.1 fm

Čopič VI/2



- 1) $\vec{E}(\vec{r}) = ?$
- 2) $\varphi(\vec{r}) = ? \rightarrow \vec{E} = -\nabla\varphi$
- 3) $E_{el} = \frac{1}{2} \int \varphi(\vec{r}) d\epsilon$

$$\epsilon_0 \oint \vec{E}(\vec{r}) \cdot d\vec{S} = e(\vec{r})$$

$\hookrightarrow$ sferna simetrija $\Rightarrow \vec{E} \neq \vec{E}(\theta, \varphi); \vec{E} = E \hat{e}_r$

i) znotraj jedra

$$\epsilon_0 \cdot 4\pi r^2 E_z(r) = e(r) = \rho_e V(r) = e_0 \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi R^3} \rightarrow E_z(r) = \frac{e_0 r}{4\pi \epsilon_0 R^3}$$

ii) zunaj jedra

$$\epsilon_0 \cdot 4\pi r^2 E_z(r) = e_0 \rightarrow E_z(r) = \frac{e_0}{4\pi \epsilon_0 r^2}$$

$$\nabla = \frac{\partial}{\partial r} \hat{e}_r + \dots \quad \leftarrow \text{ne odv. od } \theta, \varphi$$

$$E(r) = -\frac{\partial}{\partial r} \varphi(r) \rightarrow \varphi(r) = -\int E(r) dr$$

i) znotraj jedra

$$\varphi_z(r) = -\int \frac{e_0 r dr}{4\pi \epsilon_0 R^3} = -\frac{e_0 r^2}{8\pi \epsilon_0 R^3} + C_1$$

$$\rightarrow \varphi(r \rightarrow \infty) = 0, \quad \varphi_z(r=R) = \varphi_z(r=R)$$

$$\downarrow$$

 $C_2 = 0$

$$\downarrow$$

 $C_1 = \frac{3}{8} \frac{e_0}{\pi \epsilon_0 R}$

ii) zunaj jedra

$$\varphi_z = -\int \frac{e_0}{4\pi \epsilon_0 r^2} dr = \frac{e_0}{4\pi \epsilon_0 r} + C_2$$

$$\rightarrow \varphi(r=0) = C_1 = \frac{3}{8} \frac{e_0}{\pi \epsilon_0 R}$$

$$E_d = \frac{1}{2} \int \psi_c(r) dr = \frac{3}{2} \frac{e_0}{R^3} \frac{e_0}{8\pi\epsilon_0 R} \int_0^R r^2 \left[3 - \left(\frac{r}{R}\right)^2 \right] dr = \frac{3e_0^2}{20\pi\epsilon_0 R} = \frac{3}{5} \frac{e_0^2}{4\pi\epsilon_0 \hbar c} \hbar c \frac{1}{R} \sim \text{MeV}$$

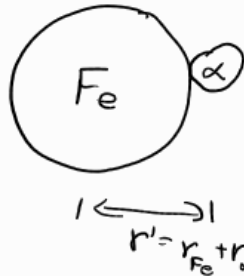
$\alpha = \frac{1}{137}$
 konst. fine
 strukture
 $\hbar c \frac{1}{R} \sim \text{MeV}$
 197 MeV fm

$$\hookrightarrow e(r) = e_0 \frac{r^3}{R^3}$$

$$\frac{de}{dr} = 3e_0 \frac{r^2}{R^3}$$

Čopić VI/3 (kok T rabi α delec, da pride do Fe jedra?)

$$\alpha = \left[{}^4_2\text{He}^{2+} \right], {}^{56}_{26}\text{Fe}$$



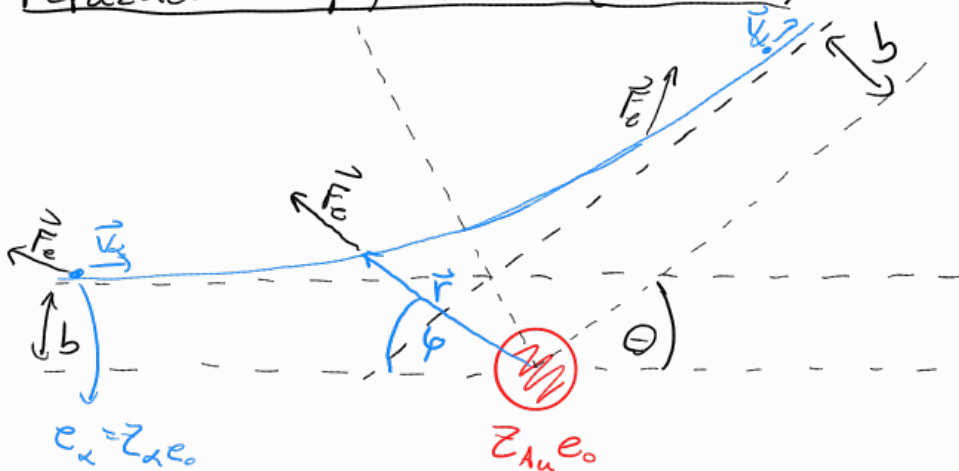
$$r_j = r_1 A^{1/3}$$

$$\rightarrow r' = r_1 (56^{1/3} + 4^{1/3}) = 5,96 \text{ fm}$$

$$E_{\alpha(\text{min})} = \psi_{Fe}(r') e_{\alpha} =$$

$$= \frac{e_0 Z_{Fe}}{4\pi\epsilon_0 r'} e_0 Z_{\alpha} = Z_{Fe} Z_{\alpha} \propto \hbar c \frac{1}{r'} = 12,5 \text{ MeV}$$

Perazdelitev sipanja α na Au (Rutherford)



$$\vec{F}_e \parallel \vec{r} \Rightarrow \vec{M} = 0 \Rightarrow \vec{L} = \text{konst.}$$

$$\vec{L} = \vec{r} \times \vec{p} = m \vec{r} \times \dot{\vec{r}}$$

$$m_{\alpha} \ll m_{Au} \Rightarrow Au \text{ je primirna, } \mu = m_{\alpha}$$

$$\vec{r} = (-r \cos \varphi, r \sin \varphi, 0)$$

$$\dot{\vec{r}} = (-\dot{r} \cos \varphi + r \sin \varphi \dot{\varphi}, \dot{r} \sin \varphi + r \cos \varphi \dot{\varphi}, 0)$$

$$\left. \begin{aligned} \vec{L} &= m(0, 0, -r \cos \varphi (\dot{r} \sin \varphi + r \cos \varphi \dot{\varphi}) \\ &\quad - r \sin \varphi (-\dot{r} \cos \varphi + r \sin \varphi \dot{\varphi})) \end{aligned} \right\}$$

$$= m(0, 0, -r^2 \dot{\varphi}) \rightarrow r^2 \dot{\varphi} = \text{konst.}$$

VK α -delca pred sipanjem: $|\vec{r}| = m|\vec{r}(t \rightarrow -\infty) \times \vec{v}(t \rightarrow -\infty)| = mrv \sin \beta = m v_0 r_t = \underline{m v_0 b}$

$$m r^2 \ddot{\varphi} = \cancel{m} v_0 b \rightarrow \ddot{\varphi} = \frac{b v_0}{r^2}$$

NZ v y-smeri:

$$m \frac{d v_y}{d t} = F_y$$

$$m \frac{d v_y}{d \varphi} \frac{d \varphi}{d t} = \frac{e^2 Z_\alpha Z_{Au}}{4 \pi \epsilon_0 r^2} \sin \varphi$$

$$d v_y = \frac{e^2 Z_\alpha Z_{Au}}{4 \pi \epsilon_0 r^2 m} \frac{\sin \varphi}{\dot{\varphi}} d \varphi = \frac{e^2 Z_\alpha Z_{Au}}{4 \pi \epsilon_0 m v_0 b} \sin \varphi d \varphi$$

$$v_y(\varphi) = \frac{e^2 Z_\alpha Z_{Au}}{4 \pi \epsilon_0 m v_0 b} (1 - \cos \varphi)$$

$$t \rightarrow \infty: \varphi = \pi - \theta$$

$$v_y(\theta) = \frac{e^2 Z_\alpha Z_{Au}}{4 \pi \epsilon_0 m v_0 b} (1 + \cos \theta) \rightarrow v_y = v_0 \sin \theta \rightarrow b = \frac{e^2 Z_\alpha Z_{Au}}{4 \pi \epsilon_0 m v_0^2} \frac{1 + \cos \theta}{\sin \theta} = \frac{C}{\tan \frac{\theta}{2}}$$

α delci

$$W_k = 3 \text{ MeV}$$

$$p = 0,9997 \text{ (prepuščnost)}$$

$$\beta = 19,3 \frac{\text{g}}{\text{cm}^3}$$

$$^{197}_{79} \text{Au}$$

$$d = ?$$

$$\frac{d \sigma}{d \cos \theta} = 2 \pi \frac{C^2}{b} \frac{1}{\sin^4 \frac{\theta}{2}}$$

$$\sigma = \left| \frac{\pi C^2}{2} \int_0^{-1} \frac{d \cos \theta}{4(1 - \cos \theta)^2} \right| =$$

$$= \left| -\frac{\pi C^2}{2} \int_1^2 \frac{du}{4u^2} \right| = \left| -\frac{\pi C^2}{8} \left(-\frac{1}{u} \right) \right|_1^2 = \left| \frac{\pi C^2}{8} \left[\frac{1}{2} - 1 \right] \right| = \frac{\pi C^2}{16}$$

$$N_r = N_v \cdot N_+ \cdot \frac{6}{5}$$

st. vpadnih st. v tarči

$$\frac{N_r}{N_v} = 1 - p$$

$$N_t = n_t V = n_t S d \quad \rightarrow n_t = \frac{\rho}{m_s}$$

$$1 - p = n_t d \sigma \rightarrow d = \frac{(1 - p) m_s}{\rho \sigma} = \dots = 1,1 \mu\text{m}$$

6.4) SEMF

$$M(^{112}_{48} \text{Cd}) = 111,902758 u$$

$$M(^{113}_{48} \text{Cd}) = 112,904400 u$$

$$M(^{114}_{48} \text{Cd}) = 113,903357 u$$

$$u = 931,484 \text{ MeV}/c^2$$

$$W_k = ?$$

$$M(A, Z) c^2 = (A - Z) m_n c^2 + Z m_p c^2 - w_0 A + w_1 A^{2/3} + w_2 \frac{Z^2}{A^{1/3}} + w_3 \frac{(A - 2Z)^2}{A} + w_4 \frac{\sqrt{Z N}}{A^{3/4}}$$

$$A \rightarrow A \pm 1; A \gg 1 \Rightarrow \text{razvoj}$$

$$w_1: A^{2/3} \rightarrow (A \pm 1)^{2/3} = A^{2/3} \left(1 \pm \frac{1}{A} \right)^{2/3} \approx A^{2/3} \left(1 \pm \frac{2}{3} \frac{1}{A} \right)$$

$$w_2: A^{-1/3} \rightarrow A^{-1/3} \left(1 \pm \frac{1}{3} \frac{1}{A} \right)$$

⋮

$$M(A \pm 1, Z) = (A \pm 1 - Z) m_n c^2 + Z m_p c^2 - w_0 (A \pm 1) + w_1 A^{2/3} \left(1 \pm \frac{2}{3} \frac{1}{A} \right) +$$

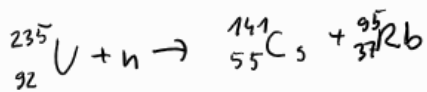
$$+ w_2 \frac{Z^2}{A^{1/3}} \left(1 \pm \frac{1}{3} \frac{1}{A} \right) + w_3 \left[\frac{(A - 2Z)^2}{A} \pm \left(1 - 4 \frac{Z^2}{A^2} \right) \right] + w_4 A^{-3/4} \left(1 \pm \frac{3}{4} \frac{1}{A} \right) \sqrt{Z N}$$

$$\rightarrow [M(A+1, Z) + M(A-1, Z)] c^2 - 2 \left[(A-Z) m_n c^2 + Z M_H c^2 - w_0 A + w_1 A^{2/3} + w_2 \frac{Z^2}{A^{1/3}} + w_3 \frac{(A-2Z)^2}{A} \right] +$$

$$\rightarrow w_4 = - \left[\frac{1}{2} (M(A+1, Z) + M(A-1, Z)) - M(A, Z) \right] c^2 A^{3/4} \quad \left| \begin{array}{l} M(A, Z) c^2 \\ (\sigma_{Z, N} = 0) \end{array} \right. \quad + \frac{2w_4}{A^{3/4}} \underbrace{\sigma_{Z, N+1}}_{=-1} \quad (Z, N \pm 1 \text{ vsi sodi})$$

$$= \underline{\underline{43.3 \text{ MeV}}}$$

Čopić VI/6



• zanemarimo T

a) ocena za spr. E (SEMF):

$$Q = E_{\nu, k} - E_{\nu, z} = E_{\nu}(\text{Cs}) + E_{\nu}(\text{Rb}) - E_{\nu}(\text{U}) = \dots (\text{SEMF}) \dots = -179.7 \text{ MeV}$$

- $w_0 = 15.6 \text{ MeV}$
- $w_1 = 17.3 \text{ MeV}$
- $w_2 = 0.70 \text{ MeV}$
- $w_3 = 23.3 \text{ MeV}$
- $w_4 = 33.5 \text{ MeV}$

b) $p_z c = 500 \text{ MeV}$

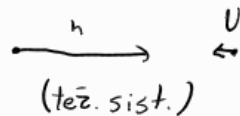
$$p' c = ?$$

$$T_{\text{Cs}} = ?$$

$$T_{\text{Rb}} = ?$$

(relativnost!)

zač.



$$\left. \begin{array}{l} p_n^{\mu} c = (E_n, p_z c) \\ p_U^{\mu} c = (E_U, p_z c) \end{array} \right\} p_z^{\mu} c = (E^*, 0)$$

$$E^* = E_n + E_U = \sqrt{(m_n c^2)^2 + (p_z c)^2} + \sqrt{(m_U c^2)^2 + (p_z c)^2} = 220 \text{ GeV}$$

koniec:

$$p_{\text{Cs}} = p_{\text{Rb}} = p'$$

$$\leftarrow \text{Rb} \quad \text{Cs} \rightarrow \quad E^* = E_{\text{Cs}} + E_{\text{Rb}} = m_{\text{Cs}} c^2 + T_{\text{Cs}} + m_{\text{Rb}} c^2 + T_{\text{Rb}}$$

$$\rightarrow T_{\text{Cs}} + T_{\text{Rb}} = 300 \text{ MeV} = \left| \begin{array}{l} T = E - m c^2 = \sqrt{(m c^2)^2 + (p c)^2} - m c^2 = \\ = m c^2 \left[\sqrt{1 + \left(\frac{p}{m c} \right)^2} - 1 \right] \approx \frac{p^2}{2m} \end{array} \right.$$

$$\rightarrow p' c = \sqrt{\frac{2(T_{\text{Cs}} + T_{\text{Rb}})}{1/m_{\text{Cs}} c^2 + 1/m_{\text{Rb}} c^2}} = 5.63 \text{ GeV} \rightarrow \begin{array}{l} T_{\text{Cs}} = 121 \text{ MeV} \\ T_{\text{Rb}} = 179 \text{ MeV} \end{array}$$

KLASIČNA
LIMITA

3) Lab. sist. z mirujočin U

$$E_n = ?$$

$$p_n c = ?$$

$$\left. \begin{array}{l} p_0^{\mu} c = (E_0, 0) \\ p_n^{\mu} c = (E_n, p_n c) \end{array} \right\} p_{\text{Lab}}^{\mu} c = (E_0 + E_n, p_n c)$$

$$E^{*2} = (E_0 + E_n)^2 - (p_n c)^2 = E_n^2 + 2E_0 E_n + E_0^2 - (p_n c)^2 =$$

$$= (m_n c^2)^2 + \cancel{(p_n c)^2} + 2m_0 c^2 \sqrt{(m_n c^2)^2 + (p_n c)^2} + (m_0 c^2)^2 - \cancel{(p_n c)^2}$$

$$\rightarrow p_n c = \sqrt{\left[\frac{(m_n c^2)^2 + (m_0 c^2)^2}{2m_0 c^2} \right]^2 - (m_n c^2)^2} = 0,49 \text{ GeV} ; E_n = 1,06 \text{ GeV}$$